

```
1  #include "FibonacciSequence.h"
2
3  FibonacciSequence::FibonacciSequence() noexcept :
4      fPrevious(0), fCurrent(1)
5  {
6
7  }
8
9  const uint64_t& FibonacciSequence::operator*() const noexcept
10 {
11     return fCurrent;
12 }
13
14 FibonacciSequence& FibonacciSequence::operator++() noexcept
15 {
16
17     fCurrent = fCurrent + fPrevious;
18     fPrevious = fCurrent - fPrevious;
19
20     return *this;
21 }
22
23 FibonacciSequence FibonacciSequence::operator++(int) noexcept
24 {
25     FibonacciSequence temp = *this;
26     ++(*this);
27     return temp;
28 }
29 }
30
31 bool FibonacciSequence::operator==(const FibonacciSequence& aOther) const noexcept
32 {
33     return fPrevious == aOther.fPrevious && fCurrent == aOther.fCurrent;
34 }
35 }
36
37 void FibonacciSequence::begin() noexcept
38 {
39     fPrevious = 0;
40     fCurrent = 1;
41 }
42 }
43
44 void FibonacciSequence::end() noexcept
45 {
46
47     fPrevious = 1;
48     fCurrent = 0;
```

49

50 }